ARC243FinalExam-DASHALTANKHUYAG

Required Question:

Design for fire protection seems to highlight the growing rift between the architect as project visionary and coordinator and the ever-increasing complexity of building systems. Can an architect properly design a fire-safe building without a detailed understanding of mechanical and electrical fire protection systems? If yes, how so? If not, what's the solution?

Answer:

Design for fire protection is a complex and ever-evolving field that requires a deep understanding of building codes, fire behavior, and fire protection systems. Architects play a critical role in fire protection by designing buildings that are safe for occupants and that minimize the risk of fire damage. *"There are four common design intents related to building fire safety in order of usual importance, a. Protection of life, protection of the building, protection of content, and continuity of operation." "Quote text (p.1143 Chapter 25)".*

"Many of the elements of building fire safety are covered by the building code, but it is important to remember that the code specifies the minimum acceptable fire protection." "Quote text (p.1143 Chapter 25)". However, the increasing complexity of building systems has made it more difficult for architects to design fire-safe buildings without the assistance of specialized fire protection engineers and or fire code enforcement authorities *"Quote text (p.1144 Chapter 25)".*It's also important that codes can't cover all aspects of fire safety there are so many variables in building design.

1st solution:

Architects can design fire-safe buildings by collaborating with specialized fire protection engineers and fire code enforcement authorities early in the design process.

"Other environmental control systems for fire safety – the desirable design approaches for fire safety will track design approaches for lighting, thermal mass, acoustic, and water systems.

The biggest challenge facing architects is the increasing use of combustible materials in the building. It is making it more difficult to evacuate occupants and extinguish fires. Design fire protection involves a comprehensive approach that fire prevention, passive fire protection, and active protection systems. *"A reasonable goal is the evacuation of all occupants at the time interval. Designers can provide clearly defined pathways to exit and provide a minimum clear width of the wheelchair. (exit access) "Quote text (p.1146 Chapter 25)"*

2nd solution:

Architects play a crucial role in incorporating fire alarm systems into building designs to ensure the safety of occupants in the event of a fire. Fire alarm systems are essential components of fire protection systems, providing early detection and warning of fires to enable timely evacuation and emergency response. *"The fire alarm system can be classified according to location, application, connections, coding, and the degree of automation of the detection system, and the classification system found in the National Fire Alarm and Signaling NFPA72 "(p.1191 Chapter 25)".* (household fire warning system, b.Protected-premises fire alarm system, c. Remote station protective signaling system, f. central station fire alarm system). Architects can collaborate with fire protection engineers to select the appropriate fire alarm system components and ensure their proper placement and integration with other fire protection systems.

I am personally interested in automatic fire detection systems, architects can design fire detection systems for buildings, especially in a single dwelling house because the detectors are best to detect fire in the early stage of the fire. Architects can make decisions about detector placement, coordination with building systems, and fire code compliance with fire authorities. *Fire authorities agree that most fires pass through four stages in Fig 25.40 p.1198 as a function of duration and degree of hazard " (p.1197 Chapter 25)".*

Conclusion

In conclusion, fire safety design demands a comprehensive understanding of building codes, fire behavior, and fire protection systems. Architects play an important role in safeguarding lives and minimizing property damage by integrating fire protection measures into their designs.

Collaborative efforts with fire protection specialists and fire code enforcement authorities are crucial to ensure that the latest fire safety principles and code requirements are seamlessly integrated into the design process.

QUESTION 04 – CHAPTER 20 15-POINTS

By today's thinking, mixing stormwater and sanitary drainage in a single pipe/system seems a less than brilliant idea – yet this was the norm and recommended practice only 100 years ago. What has happened in the past 50 or so years to change attitudes and practices in this area?

Answer:

Stormwater: Best Practice for Stormwater

"The stormwater runoff from building, and paved surfaces was considered harmless and was conducted without treatment, and rapidly as possible to receive rivers, and lakes, watersheds, basins, etc., (p.989 Chapter 20)".

Photo reference:

Project location: 6512 36th street, Arlington VA – As for me, I worked on this project and collaborated with 3rd party SW Engineer to install stormwater facilities for this house.

It was a great experience learning about stormwater facilities and implementing good stormwater practices for this project under the County's SW regulation.

Arlington County LDA 2.0 Stormwater Regulation / BMP adaption for a single dwelling – After completing SW facility installation, I understood of the environmental impacts of stormwater runoff and the need to protect the water source.

So Sanitary drainage can't connect in a single pipe with Stormwater, known as sewage disposal is the process of collecting, and draining the wastewater from the house to the main line. The reason that it is a crucial component of public health, and environmental protection is to prevent the spread of disease and contamination of water resources.

Sewage sumps and ejectors – Whenever subsoil drainage, fixtures, or equipment are situated below the level of a public sewer, A sump pit or receptacle must be installed. The drainage from low fixtures may flow into this pit by gravity. *(p.942 Chapter 20).*

Reference Photo: 6258 N Kensington street, Mclean VA

In this project, the plumbing subcontractor installed ejector sump pump and crock, Pump is motor motor-driven centrifugal pump. Reason that of manhole / public sewage line is higher than basement floor that is reason ejector pump installed, and sewer line connection method is hung-sewer.

Clean-out - clogging of sanitary drainage pipe system is very possible, From all plumbing fixtures to end of the disposal process, all parts of a sanitary piping system should be accessible through cleanouts. *(p.943 Chapter 20). As for this project, the clean-out pipe is at the front of the house.*

Conclusion:

The separate stormwater and sanitary drainage systems have been driven by a growing understanding of the environmental and public health risks associated with this outdated practice. Environmental regulations, increased public awareness, and emerging technologies have all contributed to this positive change.

QUESTION 02 – CHAPTER 18 15-POINTS Do you feel that appropriate use of water in buildings has received as much attention in discussions of "green" or" sustainable" building design as have energy and materials issues? If not, why not?

Answer:

1.
The answer is "No", I don't believe that appropriate water use in buildings has received as much attention "Green" or "Sustanable" building design as have energy and materials. Water efficiency and conservation standards are not nearly as extensive as energy efficiency standards. (p.38 Chapter 2) However, Renewable energy source, green building design efforts have also increased and design for water supply savings and alternatives, "Design approaches" for water and waste that would likely be used in a green building. (p.38 Chapter 2) water is often perceived as a readily available resource, especially in regions with plentiful rainfall. This can lead to a sense of complacency and overlooking water conservation as a critical aspect of sustainability.

2.
Water regulations and Usage of water in buildings

In my understanding, many people are unaware of the significant environmental impact of building water usage. Wastewater treatment management and the entire water cycle carry an energy cost. While energy and material-related building codes are becoming increasingly stringent, water conservation often lags behind, offering less incentive and guidance for building developers and designers. However, civil engineers can play a critical role in designing sites, calculating rainwater runoff, and stormwater management, water conversation and preserving tree areas. Nonetheless, there is a lack of specific design guidelines and standards concerning water usage in buildings.

The materials selection has been the cornerstone of the green building efforts – which offer tangible and measurable benefits. A quantitative comparison with water must suffice. Many materials used for building construction and repairs come from a generally fixed source base and are renewable (p.39 Chapter 2).

Nonrenewable materials are by far the most commonly used constituents of the mechanical and electrical system, metal, and plastics predominate. Their advantage include strength, durability, fire resistance, and conductivity or resistivity as required. Increasing demand for both renewable or nonrenewable materials depends on how a designer use the materials. (p.39 Chapter 2)

Conclusion:

Water conservation has not received as much attention as energy and materials in sustainable building design. However, there is a lack of specific design guidelines and standards concerning water usage in buildings.

QUESTION 08 – CHAPTER 25 15-POINTS What parties to the building design and construction process—in addition to architects, mechanical engineers, and electrical engineers—will play a substantive role in establishing the success of fire protection systems? For each party that is identified, briefly outline the role they can/should play.

1. Contractors (fire protection system contractors and installer:
    a. Install the sprinkler systems, fire alarm systems, and other fire systems installation according to design specifications and codes.
2. County fire code officials and fire marshals:
    a. Review fire design documents and inspect installed fire systems for residential and commercial. *"Sprinkler system design – sprinkler systems are widely relied on as proven automatic fire suppressers. However, fire designers must remember that the use of sprinklers doesn't give one a license to ignore fire code limitations, even though many codes provide more for sprinklers (p.1161 Chapter 25).*
    b. *Fire designers through fire safety codes and regulations.*
3. Fire protection engineer:
    a. Perform fire risk assessments, develop fire protection strategies, and design sprinkler systems, specify fire alarm and detection systems, and ensure code compliance.
4. Firefighters
    a. Respond to fire emergencies, and utilize fire protection systems effectively

QUESTION 09 – CHAPTER 26 15-POINTS

The authors state that "planners must be familiar with normal electrical systems." Assuming the "planners" are architects, do you believe that this familiarity is in fact the case for most architects? If so, is this familiarity attained in school; in practice? If not, why not? If not, how do you think this affects the design of buildings?

Answer:

The authors state that "planners must be familiar with normal electrical systems." Assuming the "planners" are architects, do you believe that this familiarity is the case for most architects?

I think it is difficult to give a definitive answer that a planner must be familiar with normal electrical systems, however, it could be due to (a) education and (b) experience working on projects requiring electrical components to gain practical knowledge through collaboration with electrical contractors and or engineer. I do believe that most planners and or architects are familiar with designing principles of electricity regarding their experience and electrical codes. I felt that architects have a limited understanding of electrical systems because architects prioritize design, leaving the technical aspects like electrical systems. Because architects rely on electrical engineers for detailed design and implementation.

If not, how do you think this affects the design of buildings?

It does not affect the design of buildings, Architects can design principles of the electrical components like lights, showing appliance locations, receptacles and exterior electrical fixtures location, ceiling fans, and indicating the location of the electrical panel. I have experienced it in the residential new home project and filled out the electrical load form including – the electrical heat pump, all appliances, lights, and electrical components for Dominion Energy - The electrical designer helped me to design the electric – their design is detailed and specified each of the electrical components like transformer, electrical meter, electrical base size, showing all electrical wiring, etc., Then electrical contractor implemented the projects based on electrical engineer's design.

Conclusion:

The projects demonstrate how architects can contribute to the design of electrical components, it's crucial to acknowledge the limitations of relying solely on architects' knowledge. Collaboration with electrical engineers remains essential for ensuring the technical feasibility, safety, and optimal performance of electrical systems in all types of buildings.

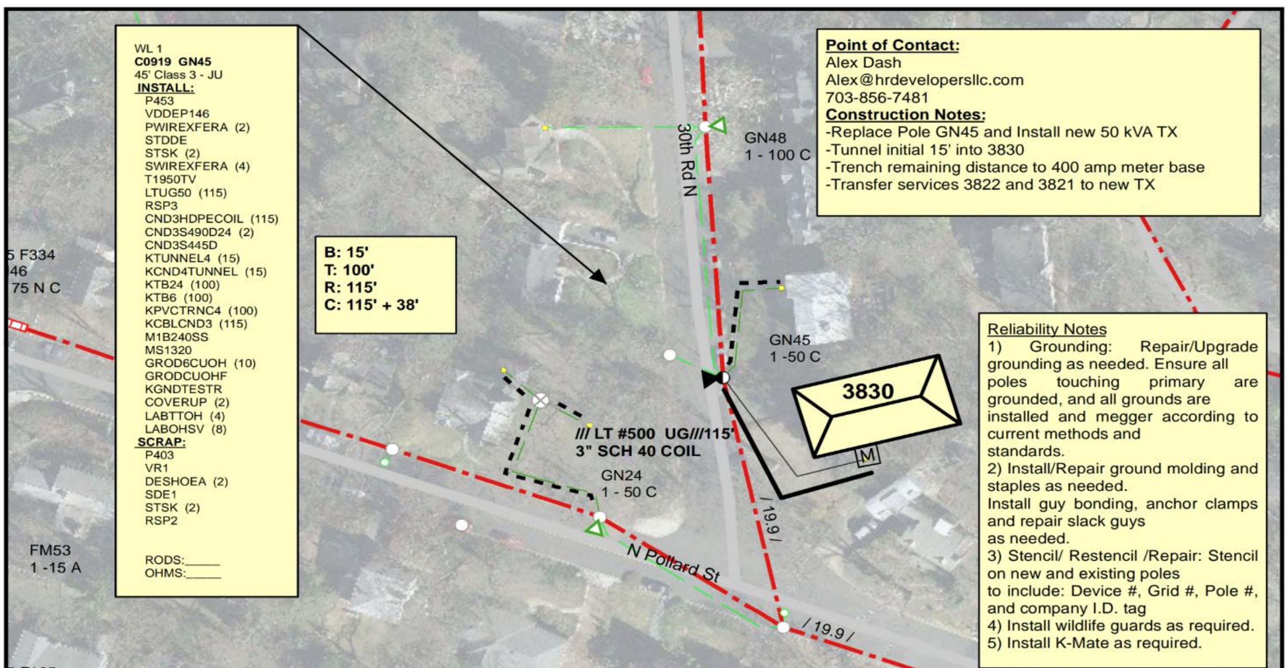
QUESTION 10 – CHAPTER 27 15-POINTS An electrical distribution system is similar in concept to other buildings distribution systems (such as conditioned air or potable water/drainage) that form a distribution tree. Similar in concept; but different in detail. Identify four important differences—from a general building design perspective—between an electric distribution tree and a conditioned air distribution tree.

Answer:

First: What is an electrical distribution system?

Distribution line from the main power line to the buildings:

"Electrical service is normally connected to the utility line at a mutually agreeable point or beyond the property line. The service tap may be a connection on a pole with an overhead service drop or an underground service lateral to the building" (p.1239 Chapter 20). The electrical distribution tree is a network of wires and cables that transports electrical power from the utility grid to various points of use within a building. It resembles



Work Description: -Replace Pole GN45 and Install new 50 kVA TX
-Tunnel initial 15' into 3830
-Trench remaining distance to 400 amp meter base
-Transfer services 3822 and 3821 to new TX

Location: 3830 30th Road N Arlington, VA 22207

Volts:120/240	Amps: 400	Phase: C	Voltage Drop: 1%	Flicker: 4.63%	Fault Current: 7437 amps	Est. Demand: 37.5 kVA

Facility ID: C0919GN4500	TRS #: 41755 F334	Circuit #: 41755	Substation: Clarendon
Easement #: N/A	Tax District: 33000	Premise ID: 0773630100	Scale: **1 inch = 74 feet**

Work Request # 10634725	Designer: Thomas L Harding	EID: THO0518	Phone #: 571-319-1192	Date: 8/7/2023

a tree-like structure, with the main power supply serving as the trunk and branch circuits extending to individual outlets, lighting fixtures, and appliances.

The service entrance: the public /main utility grid connects to the building's electrical system. It typically includes a transformer, which reduces the high-voltage power from the grid to a usable level for the building.

Reference photo: 3830 30th rd, Arlington VA – I worked with Dominion Energy to connect the power service to new house. This work description form from Dominion Energy – contains a lots information about electrical service:

Example:

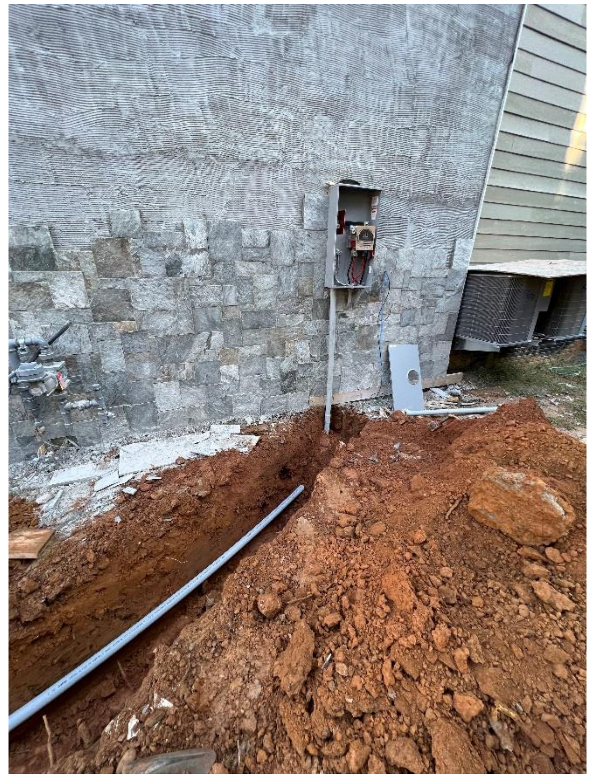
- Power pole – It depends on DE to replace the power pole and or not.
- Transformer model #: 50 kVA TX
- Tunnel initially 15' long from pole to the exterior electrical panel (M) is meaning that electrical meter location.
- Electrical meter base is usually 400 amp meter base for residential. (Volts 120/240 Amps 400 Phase C)
- Service is underground – installed 4" PVC conduit

Transformer model 50 kVA TX

The designation "50 kVA TX" refers to a transformer with a power rating of 50 kilovolt-amperes (kVA). This means that the transformer can transfer 50,000 watts of power from one voltage level to another. The "TX" stands for "transformer," indicating that this is a device that converts electrical energy between different voltage levels.

An electrical distribution system is a complex network of interconnected components that work together to safely and efficiently deliver electrical power from the utility grid to various points of use within a building.

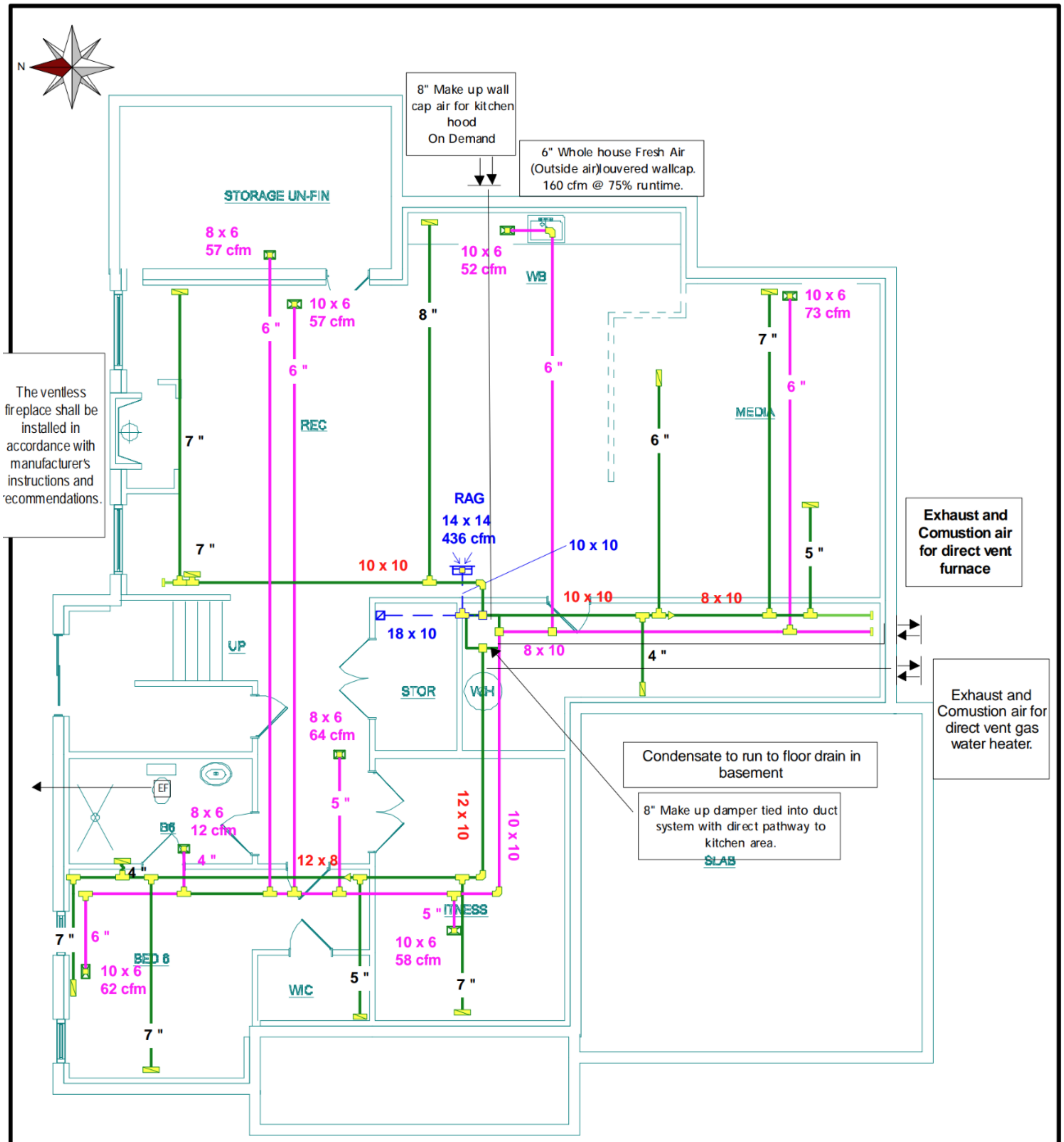
It consists of essential elements like the service entrance, main distribution panel, branch circuits, wires and cables, and outlets and switches. Each component plays a crucial role in ensuring the proper distribution of electrical power, maintaining safety, and supporting various electrical demands within the building structure.

A conditioned air distribution tree is a system of ducts and air handlers that delivers conditioned air (heated or cooled) to various zones within a building. The system is designed to maintain a comfortable indoor environment by regulating temperature, humidity, and air quality.

Conclusion:

While both electrical distribution trees and conditioned air distribution trees are critical components of building infrastructure, they differ significantly in their purpose, medium of transport, flow direction, and branching strategy, each tailored to fulfill their respective functions in delivering electricity and maintaining a comfortable indoor environment.



Job #: 3830 30TH RD N Performed for: 3830 30TH RD N (HR Dev) 3830 30TH RD N Arlington, VA	HR Developers	Scale: 1/8" = 1'0" Page 1 Right-Suite® Universal 2022 22.0.05 RSU21761 2022-Dec-01 18:58:41 ... 30TH RD N (3716 N Nelson St.rup